This thesis focuses on simulating code of Arduino microcomputers in a virtual environment with a graphical user interface without the need of emulation. An important part of the thesis shows how and why should be the simulator and the GUI split into two processes, and how to keep them synchronized. This introduces the reader into the field of interprocess communication and its implementations using TCP connection, shared memory and events. Another important part focuses on simulation of digital circuits with extensible component library. It describes how new components and circuits can be created by the user from YAML, SVG and C# files.